Antenna mast Specification

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Revision	B
Status	APPROVED
Classification	INTERNAL
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Approved by	Hesker Doyle
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1. Purpose

This specification defines the performance, interface and verification requirements for the antenna mast (SF-ANM) of the Superstructure & Frame on Starhaven Orbital Habitat & Shipyard.

2. Scope

Fabrication and installation are performed under work package WP-SHV-SF-05 (WBS 1.14.5), owned by Ferris Sable.

The prime supplier for flight hardware is Kuat Drive Yards. Responsible engineer: Edda Parrish.

3. Applicable documents

Document	Title
SHV-SYS-SPEC-0001	Starhaven Master System Specification
SHV-SF-DWG-0204	Antenna mast general arrangement drawing
SHV-SF-BOM-0304	Antenna mast bill of materials
SHV-QA-PLN-0004	Starhaven Verification and Test Plan
SHV-QA-STD-0002	Station Workmanship Standard
SHV-ICD-SF-HB	Interface Control Document: Superstructure & Frame / Docking & Hangar Bays
SHV-ICD-SF-SY	Interface Control Document: Superstructure & Frame / Shipyard Slipways
SHV-ICD-SF-TR	Interface Control Document: Superstructure & Frame / Internal Transit

4. Description

The superstructure is the station's load-bearing frame: structural ribs, hull plating sections, the equatorial service ring and the antenna masts.

The antenna mast is delivered as a set of line-replaceable assemblies, integrated on station by Construction Operations and verified by Quality Assurance & Test.

5. Performance requirements

ID	Requirement	Value	Tolerance	Verification
SF-ANM-R01	The antenna mast shall provide a mast height of 531 m.	531 m	±1%	Demonstration

5.1 General requirements

ID	Requirement	Verification
SF-ANM-G01	Maintainability: Line-replaceable units shall be replaceable by a crew of four within one shift.	Inspection
SF-ANM-G02	Safety: Single faults shall not cause loss of the zone or injury to crew.	Analysis
SF-ANM-G03	Workmanship: Welds and fasteners shall meet the station workmanship standard SHV-QA-STD-0002.	Inspection
SF-ANM-G04	Marking: Every assembly shall carry its drawing number, revision and serial number.	Analysis
SF-ANM-G05	Documentation: Acceptance data packages shall be delivered with every unit.	Inspection

6. Environmental conditions

Condition	Value
Operating pressure	Vacuum to 110 kPa
Radiation, total dose	Station design environment, class R3
Thermal cycling	200 cycles, operating range
Vibration	Station-keeping manoeuvre profile, level 2
Electromagnetic compatibility	Station EMC standard, category B

7. Verification

Each requirement is verified by the method stated in section 5, following SHV-QA-PLN-0004. Acceptance test reports are filed under the SF test-report series.

8. Revision history

Rev	Date	Description	Author
A	2022-10-14	Initial release.	Edda Parrish
B	2023-05-11	Updated requirement values after design review.	Edda Parrish